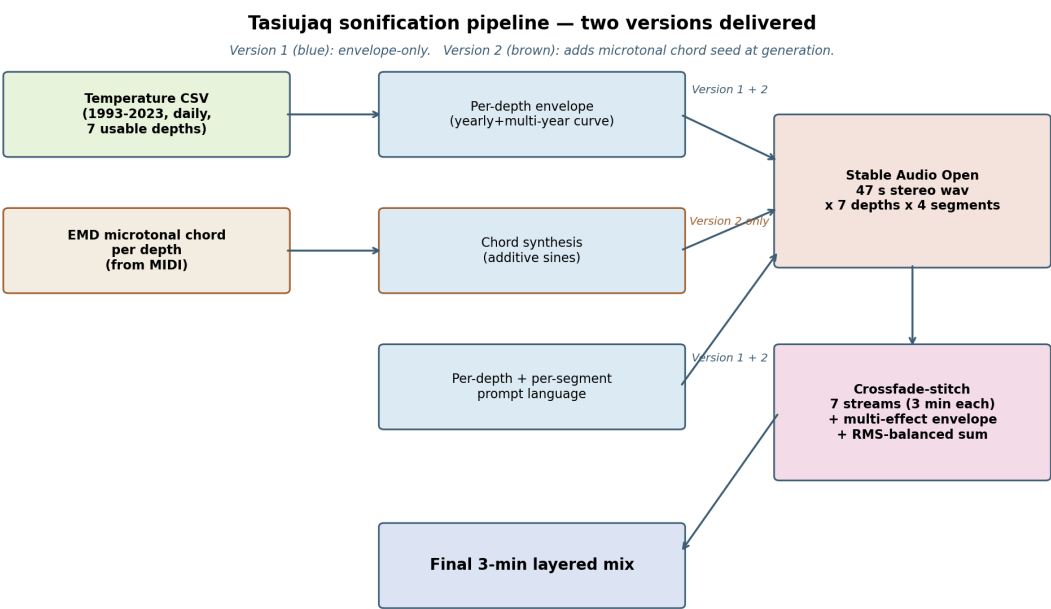


From Permafrost to Music

Two delivered sonifications of Tasiujaq soil temperature data



Summary. We turn 27 years of daily soil-temperature data from Tasiujaq (1993–2023) into two complementary 3-minute orchestral pieces. Both pieces share the same seven musical voices (one per soil depth between 200 cm and 1100 cm), the same prompt-driven instrument design, and the same envelope-shaping stage. They differ in how the data couples to the generated audio:

Version 1 — Envelope-modulated sonification. The soil signal shapes each voice via a per-depth audio effect: loudness, brightness, stereo width, tremolo, or bass cut, chosen so each voice expresses the data in a different sensory dimension.

Version 2 — Microtonal-anchored sonification. In addition to the envelope shaping, the microtonal harmonic content of each depth (extracted by EMD) is fed directly into the AI model's diffusion process as an initial seed. The model's generated pitches are then biased toward the depth's actual microtonal chord. The data couples to the music both in time (envelope) and in spectrum (chord).

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1 The source data

The input is a CSV file of soil temperatures measured at fourteen depths between 0 cm (surface) and 1100 cm (eleven metres below ground) at Tasiujaq, Nunavik. Measurements are daily and span 26.7 years (9,748 days, August 1993 through 2023). Seven of the fourteen depths have sufficiently continuous coverage to be usable in the piece: 200, 300, 400, 500, 700, 900 and 1100 cm.

The depths tell radically different physical stories. Near the surface, the soil follows the seasons aggressively — summer warming pushes temperatures up past +20 °C, while winter air drops them below −30 °C. Heat diffuses slowly into the earth, so every additional metre of depth smooths and delays this signal. By 11 m deep, temperatures hover narrowly between −5 and −4 °C all year round — that is the permafrost, almost motionless across the full 27-year record.

A complementary input is also available: an Empirical Mode Decomposition (EMD) of the same soil signal, exported as a microtonal MIDI file. For each usable depth, the EMD analysis produces 8–11 frequencies spread across roughly eight octaves — a per-depth “chord” that characterises the depth’s internal harmonic structure. The lowest of these frequencies is always around 28–32 Hz; this shared sub-bass anchor reflects the multi-decade temperature trend that every depth ultimately responds to.

┃ *Volatile soil → expressive voice. Stable soil → drone.*

2 From depth to voice

Each of the seven usable depths is assigned its own musical voice — an instrument or vocal layer described in prose and fed to the AI model as a text prompt. The voices are deliberately spread across distinct frequency bands and roles so they layer transparently rather than crowd the same range.

Per-depth voice assignment			
Depth	Physical character	Musical voice (v10)	Role
200 cm	Volatile (std ~ 3.4 °C)	Soft soprano + celesta	Melodic top voice
300 cm	Volatile (std ~ 2.1 °C)	Piano + warm Rhodes	Melodic mid voice
400 cm	Moderate (std ~ 1.2 °C)	Muted viola + humming	Lead lament
500 cm	Moderate (std ~ 1.1 °C)	Wordless choir	Harmonic pad
700 cm	Stable (std ~ 0.7 °C)	Taiko + pizzicato cello	Rhythmic build
900 cm	Stable (std ~ 0.36 °C)	Cello section + male bass	Low foundation
1100 cm	Near-frozen (std ~ 0.36 °C)	Permafrost + throat singing	Sub-bass drone

Figure 1. Per-depth voice assignment, shared by both versions. Warm tones at the top are the most expressive and melodic; cool tones at the bottom hold the drone foundation. Surface stems contain the more delicate, transient elements (voice, celesta); deep stems are sustained and in the low register.

3 Mapping 27 years onto 3 minutes

The AI audio model’s hard architectural maximum output is 47 seconds. To produce a 3-minute piece we split the audio into four 47-second segments crossfaded together (3-second equal-power crossfades, giving $4 \times 47 - 3 \times 3 \approx 179$ s). We use this constraint to our narrative advantage: each segment is fed a different ~ 6.7 -year slice of the soil record, so the multi-decade warming trend is embedded in the music’s progression.

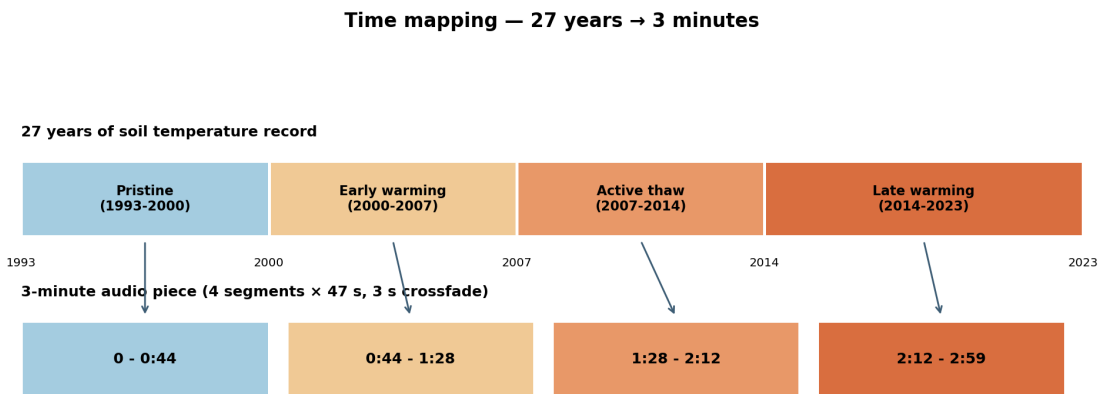


Figure 2. Time mapping. The 27-year record is divided into four windows of about 6.7 years each. Each window drives one ~ 47 -second audio segment; all four are crossfaded together into the final ~ 3 -minute piece. Prompt descriptors evolve in parallel — “pristine” at the start, “active thaw” in the middle, “late warming” at the end.

Because each 47-second segment is driven by ~ 6.7 years of daily temperature data, every segment contains roughly seven seasonal cycles of the soil signal. The envelope therefore oscillates at both a slow multi-year scale (the climate trend) and a faster yearly scale (the seasons) within each 47-second window.

4 The shared mechanism: envelope extraction

Both versions share the same first stage: for each depth and each segment, the actual temperature curve is converted into an envelope — a single curve in $[0, 1]$ that follows the soil signal in time. That envelope is what sculpts the AI-generated audio in the second stage.

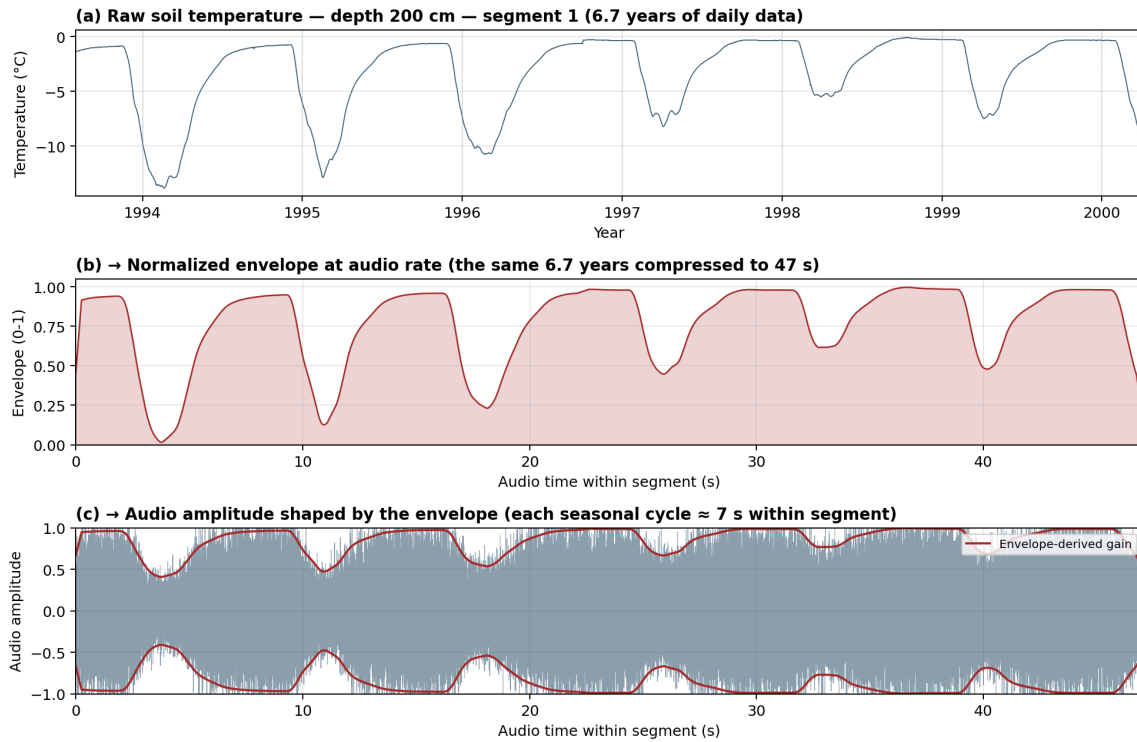


Figure 3. The three-stage transformation, depth 200 cm, first segment (1993–2000). (a) About 6.7 years of daily temperature data — seven seasonal cycles are clearly visible. Vertical dotted lines mark January 1; troughs land near January (winter), peaks land mid-year (summer). (b) The same shape is normalised to $[0, 1]$, smoothed, and resampled onto the audio time grid; each year occupies ~ 7 seconds within the 47-second segment. (c) The envelope sculpts a synthesised waveform: loudness and other effects breathe with the seasons.

The audio that the envelope shapes is produced by Stable Audio Open 1.0, an open-weights text-to-audio model run locally on a single GPU. For each combination of depth $\times$ segment we write a tailored prompt describing the voice’s instrument, register, and segment-specific mood. With seven depths and four segments per depth, the model is invoked 28 times per piece. Generation is deterministic given a fixed random seed.

5 Version 1 — Envelope-modulated sonification

Version 1 in one line. The AI generates each voice from a text prompt. The depth’s temperature envelope then drives one signature audio effect on that voice — chosen so each voice expresses the data in a different sensory dimension.

5.1 Per-depth effect routing

The amplitude envelope is always applied (loudness rises and falls with warmth). On top of that, each depth gets one additional, voice-appropriate effect. The routing is intentional: bright, melodic voices receive an effect that varies their brightness or spatial presence; foundational drones receive a low-frequency effect that lets the listener hear the bass “thaw” as the soil warms; rhythmic voices keep amplitude only, so the listener can still feel the rhythm clearly.

Depth	Voice	Envelope-driven effect (cold → warm)
200 cm	Soprano + celesta	Stereo width. Intimate ↔ wide.
300 cm	Piano + warm Rhodes	Low-pass cutoff. Muffled ↔ bright.
400 cm	Muted viola + humming	Tremolo depth. Steady ↔ trembling.
500 cm	Wordless choir	Stereo width. Narrow ↔ wide.
700 cm	Taiko + male bass voice	None (amplitude only — preserves rhythmic clarity).
900 cm	Cellos + male bass voice	High-pass cutoff. Bass cut ↔ full bass.
1100 cm	Permafrost + throat singing	None (amplitude only — subtle drone).

5.2 Implementation

All envelope-driven filters are implemented as a sample-wise crossfade between a pre-computed fully-filtered version of the signal and the dry signal — this avoids the audible transients that block-wise IIR filtering produces when the cutoff sweeps quickly. Tremolo is implemented as a low-rate LFO whose depth is modulated by the envelope (steady when cold, trembling when warm). Stereo width modulates the mid/side ratio.

The same temperature curve sculpts seven voices, each in its own way.

6 Version 2 — Microtonal-anchored sonification

Version 2 in one line. Same envelope effects as Version 1, plus: the depth’s EMD-derived microtonal chord is fed to the AI model as an initial seed of the diffusion process. The model’s generated pitches are biased toward that chord — the soil data couples to the music both in time and in spectrum.

6.1 The chord seed

For each depth, the partials extracted from the microtonal MIDI are synthesised as an additive-sine 47-second stereo clip. Partial above 2 kHz are dropped (they would inject strident high content) and the remaining partials are weighted so that the lowest frequencies dominate.

Version 2 — Microtonal chord seeding of the diffusion process

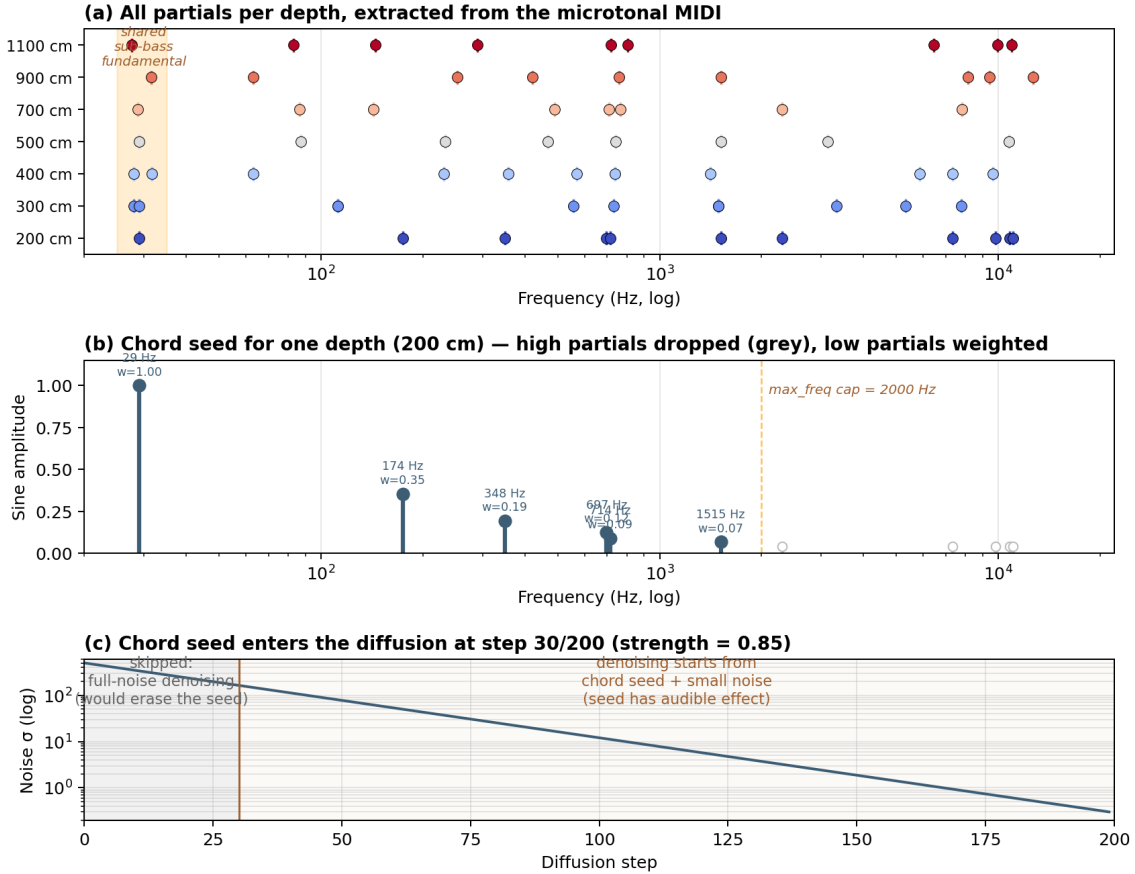


Figure 4. Construction and integration of the Version 2 chord seed. (a) The full partial set per depth, as extracted from the microtonal MIDI. The orange band marks the shared sub-bass fundamental near 30 Hz that links all seven depths into one harmonic family. (b) For one depth (200 cm), the partials kept (blue) and dropped (grey) after the high-frequency cap, with the per-partial amplitude weights used in additive synthesis. (c) Where the chord seed enters the diffusion process: instead of starting denoising from the full-noise step (which would erase the seed), we skip the first 85% of the noise schedule and begin denoising from a much smaller starting noise. The seed therefore has an audible effect on the final spectrum.

6.2 How the seed enters the diffusion

In a standard text-to-audio generation, the model starts from pure noise and gradually denoises into audio that matches the text prompt. When an initial audio is provided, the standard pipeline still adds full-magnitude noise on top of it before the first denoising step, which effectively washes out any influence of the seed. Version 2 uses an img2img-style variant: the noise added to the encoded chord seed is scaled to a much smaller level, and the denoising loop is shortened to start from that point onwards. The model now operates on a latent that

is genuinely a noisy version of the chord, and its denoising path is pulled toward the chord’s spectrum.

A single parameter, the strength, controls the trade-off: at strength = 1.0 the seed is ignored (stock behaviour); at lower values the model is increasingly committed to the seed’s spectrum. In practice we use strength ≈ 0.85 — the seed shapes the spectrum without overpowering the model’s creative content.

6.3 What is shared with Version 1

The per-depth multi-effect routing table from Section 5 is applied unchanged to Version 2’s audio. The chord seed shapes what notes the model plays; the envelope effects then shape how those notes are heard over time. The data thus couples to the music in two orthogonal dimensions.

7 Assembly: stitching and mixing

Each depth's four segments are crossfaded together (3-second equal-power crossfades) into a single ~ 3 -minute stream. The seven streams are then loudness-balanced and summed into the final mix. The same assembly stage is used for both versions.

Mix balance — per-stem gains (after RMS equalization)

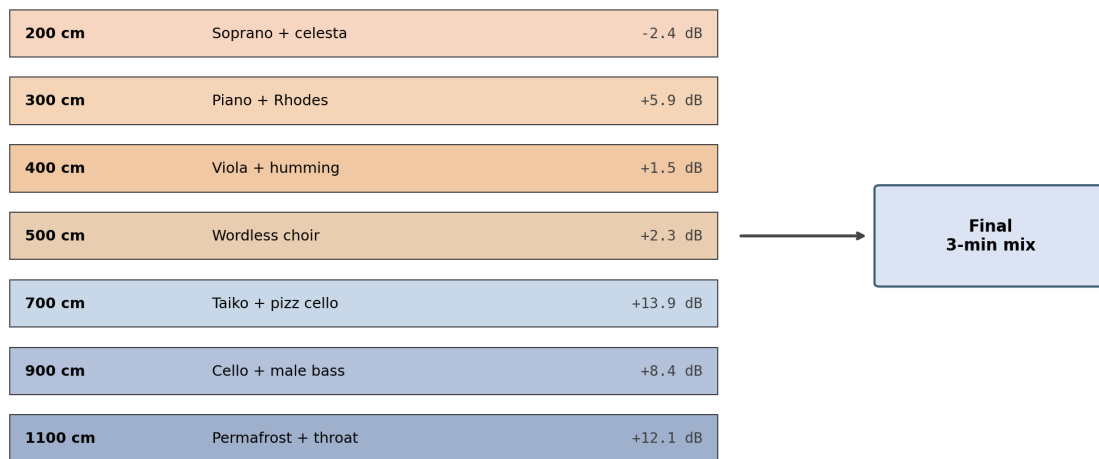


Figure 5. Mix balance. The seven streams and their final gains in the delivered mixes. Sparse, transient stems (such as the taiko drums or the permafrost sub-bass) receive larger boosts (up to +12 dB) to perceptually match the dense, sustained stems (such as the choir). The dB numbers next to each stem are the total gain applied via a two-step process: RMS equalisation (lifts sparse stems to the common loudness target) and a small manual fader adjustment per stem.

8 Roadmap

Multi-IMF envelopes — Each depth's temperature signal can itself be decomposed by EMD into a short list of frequency components (daily, seasonal, multi-year). Routing different IMFs to different effects on the same voice — e.g. the multi-year trend driving filter cutoff while the seasonal cycle drives loudness — would give three data-driven control streams per depth instead of one. No additional AI compute needed; only an iteration on the envelope-shaping stage.

Path B — envelope-conditioned generation — A more ambitious extension is to make the AI itself plan its dynamics so that the generated audio's natural loudness curve already matches the soil envelope, rather than having the envelope applied afterwards. This is achievable via inference-time optimisation of the initial noise latent (DITTO-style methods). Compute cost is roughly $10\times$ the current pipeline.

Path B⁺ — multi-target conditioning — The same inference-time optimisation can target both the envelope and the spectrum simultaneously, using the chord seed as a spectral target and the temperature curve as an envelope target. This would unify Versions 1 and 2 into a single

conditioning pass rather than a two-stage pipeline.

Provenance. Source data: Tasiujaq permafrost monitoring (1993–2023). Audio model: Stable Audio Open 1.0 (Stability AI). Pipeline implementation: this project, Python 3.12 with `diffusers`, `scipy`, `matplotlib`. Report typeset with $\text{\LaTeX}$ (Tectonic), Linux Libertine type.